INNOVATOR TEST 6
MATH MODULE 1

1. The price p for renting a boat for h hours can be found using the equation $p = 25h + 15$. If Spencer spends a total of $\$115$ to rent this boat, how many hours did he rent the boat?

NEXT QUESTION

- A 4
- B 5
- C 8
- D $2{,}890$

2. The linear function $f(x) = \frac{5}{2}x + b$ is true for all values of x. If $f(6) = 3$, what is the value of b?

PREV. QUESTION NEXT QUESTION

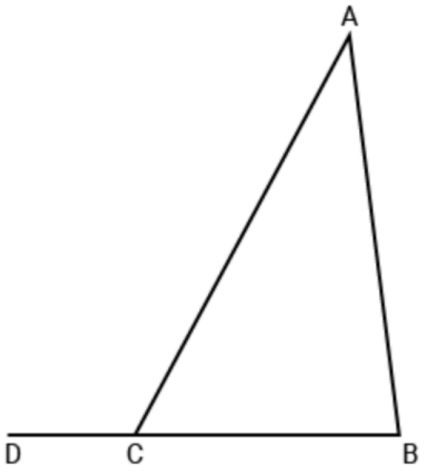
- A -12
- B -1.5
- C 12
- D 15

3. If $x = 3$, what is the value of $2\sqrt{x^4}$?

PREV. QUESTION NEXT QUESTION

COME BACK LATER

4.



In the figure shown, points B, C, and D are on the same line, the measure of $\angle A$ is 34°, and the measure of $\angle B$ is 72°. What is the measure of $\angle ACD$, in degrees?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

5.

Die Outcome	Percentage
1	20.3%
2	32.6%
3	
4	24.2%

Katy is rolling a 4-sided die and determined the percentage of each outcome. What percentage of the die rolls had an outcome of 3?

PREV. QUESTION | NEXT QUESTION

- A 20.3%
- B 22.9%
- C 23.55%
- D 25%

COME BACK LATER

6. $y = x + 1$

$y = 3x - 5$

What is the solution to the given system of equations?

PREV. QUESTION | NEXT QUESTION

- A $(-3, -4)$
- B $(-3, 4)$
- C $(3, -4)$
- D $(3, 4)$

7. What is the solution set for the inequality $2x + 5 \leq 7x + 15$?

PREV. QUESTION | NEXT QUESTION

- A $x \leq -2$
- B $x \geq -2$
- C $x \leq 10$
- D $x \geq 10$

8. Victoria gathered 10 sticks in a forest and measured their lengths in inches. The lengths of the sticks are shown.

$4.2, 6.3, 4.6, 2.0, 3.4, 7.3, 4.1, 5.7, 8.3, 9.1$

What is the mean length, in inches, of the sticks?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

9. What is the length of the hypotenuse, in feet, of a right triangle with legs with lengths of 16 feet and 30 feet?

PREV. QUESTION | NEXT QUESTION

- A 25
- B 34
- C 36
- D 46

10. $$2x^2 + 6x - 18 = 0$$

What is the sum of the solutions of the given equation?

PREV. QUESTION | NEXT QUESTION

- A -9
- B -3
- C 3
- D 9

11. $$2x - 3y = 9$$
$$ax - 5y = 3$$

For what value of a will the above system of equations have no solutions?

PREV. QUESTION | NEXT QUESTION

- A $-\frac{10}{3}$
- B $-\frac{2}{3}$
- C $\frac{2}{3}$
- D $\frac{10}{3}$

12. Which of the following is equivalent to the expression $x^3 - 4x^2 + 8x - 32$?

PREV. QUESTION | NEXT QUESTION

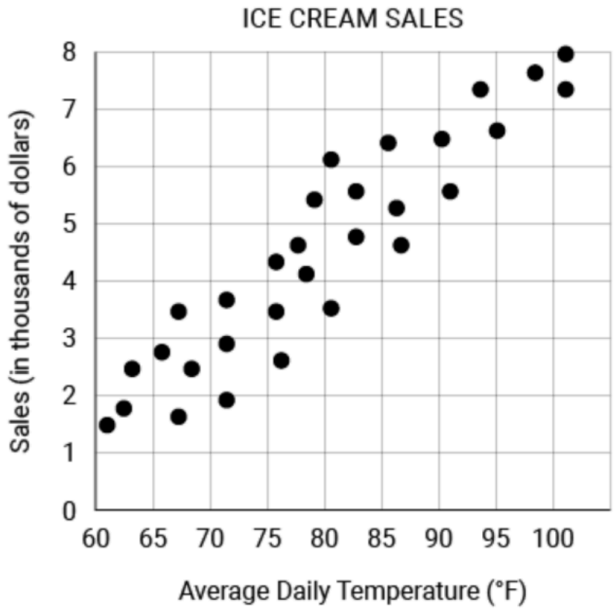
× A $(x + 8)(x - 4)$

× B $(x^2 + 8)(x - 4)$

× C $(x + 4)(x - 4)^2$

× D $(x^2 - 8)(x + 4)$

13.



The graph shows the sales of ice cream at a store and the average daily temperatures for a month. Which data point would be considered an outlier for the graph?

PREV. QUESTION | NEXT QUESTION

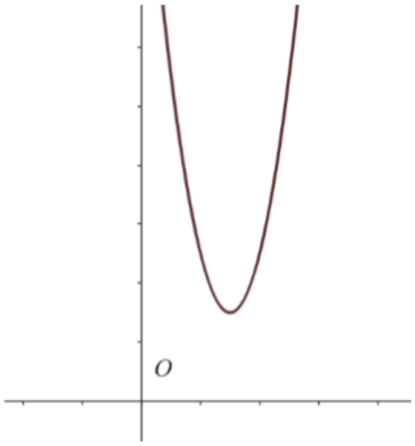
× A On a day with an average temperature of $60^\circ F$, the store makes $\$2{,}000$ in ice cream sales.

× B On a day with an average temperature of $70^\circ F$, the store makes $\$3{,}000$ in ice cream sales.

× C On a day with an average temperature of $75^\circ F$, the store makes $\$2{,}000$ in ice cream sales.

× D On a day with an average temperature of $80^\circ F$, the store makes $\$1{,}000$ in ice cream sales.

14.



For the function $y = f(x)$ shown, which of the following equations could be represented by the graph?

PREV. QUESTION | NEXT QUESTION

× A $f(x) = 2[(x - 60)^2 + 30]$

× B $f(x) = 3[(x + 15)^2 + 30]$

× C $f(x) = 5[(x - 40)^2 - 60]$

× D $f(x) = 6[(x + 50)^2 - 60]$

COME BACK LATER

15. If the equation $x(a + 2) = 12(x + 1)$ has no real solution and a is a constant, what is the value of a?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

16. A rectangular prism has a height of h inches. The width is half of the height. The length is three times the width. What is the volume of the rectangular prism?

PREV. QUESTION | NEXT QUESTION

× A $\frac{3}{4}h^3$

× B $\frac{3}{2}h^3$

× C $3h$

× D $12h^3$

17. When graphed in the xy–plane, which equation will have a slope of $\frac{5}{9}$ and an x–intercept at $x = 4$?

PREV. QUESTION | NEXT QUESTION

× A $5x + 9y = 4$

× B $5x - 9y = -36$

× C $-10x - 18y = 8$

× D $-10x + 18y = -40$

18. Which expression is equivalent to $\frac{12w^6x^{-2}y^4z}{20w^3x^{-8}y^{-12}z^3}$?

PREV. QUESTION | NEXT QUESTION

× A $\frac{3w^3x^6y^{16}}{5z^2}$

× B $\frac{3w^2y^3}{5x^4z^3}$

× C $\frac{3w^3y^{16}}{5x^6z^2}$

× D $\frac{3w^3x^{10}y^8}{5z^2}$

19.

$$y^2 + 36 = x$$

$$15y = x$$

If (x, y) is a solution to the given system of equations, which of the following is an x–coordinate of a solution?

PREV. QUESTION | NEXT QUESTION

× A 3

× B 12

× C 45

× D 175

COME BACK LATER

20. $\frac{3x}{4} + 0.325 = 3.2x - \frac{17}{8}$

What is the solution to the given equation?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

21.

x	$f(x)$	$g(x)$
−5	4	1
−2	−5	7
1	7	4
4	−2	−5
7	1	−2

The values for two functions $f(x)$ and $g(x)$ are shown in the table. If $g(a) = -5$, where a is a constant, what is the value of $f(a - 3)$?

PREV. QUESTION | NEXT QUESTION

A -2

B 1

C 4

D 7

COME BACK LATER

22. The function $f(n) = 4 + (n - 1)9$ and the function $g(n) = -2 + (n - 1)(10)$ are defined for all whole number values of n. What is the value of $f(n)$ when $f(n) = g(n)$?

PREV. QUESTION

COME BACK LATER

MATH MODULE 2 (HARDER)

1.

$$y = -2x - 3$$

$$y = x^2 - 2x - 3$$

Which ordered pair(s) is the solution to the given system of equations?

NEXT QUESTION

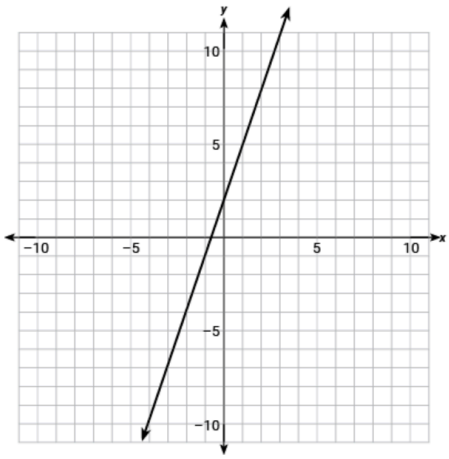
A $(0, -3)$

B $(1, -5)$

C $(3, 0)$ and $(-1, 0)$

D $(-1, -1)$ and $(-1, 0)$

2.



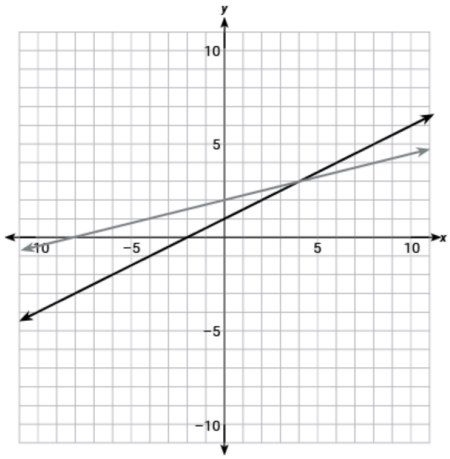
The graph of linear function $g(x)$ is shown above. What is the value of $g(2)$?

PREV. QUESTION | NEXT QUESTION

- A 0
- B 5
- C 8
- D 10

COME BACK LATER

3.



A system of linear equations is graphed above. What is the y value of the solution to the system of equations?

PREV. QUESTION | NEXT QUESTION

- A 1
- B 2
- C 3
- D 4

COME BACK LATER

4. An antique grandfather clock chimes every hour. At one o'clock it chimes once, and at two o'clock it chimes twice, and so on. Each chime rings for 1 second, and there are 3 seconds between each chime. At 8 pm, how many seconds does it take from the start of the first chime to the end of the last one?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

5. $$f(T) = \frac{9}{5}T + 32$$

The given function $f(T)$ determines the temperature in Fahrenheit, where T represents the temperature in degrees Celsius. At what temperature, in degrees, is the Celsius temperature the same as the Fahrenheit temperature?

PREV. QUESTION | NEXT QUESTION

× A -40

× B -17.8

× C 0

× D 32

6. $$3x + 2y = 8$$
$$2x - 3y = 1$$

What is the solution (x, y) to the given system of equations?

PREV. QUESTION | NEXT QUESTION

× A $(-2, -1)$

× B $(-1, -2)$

× C $(1, 2)$

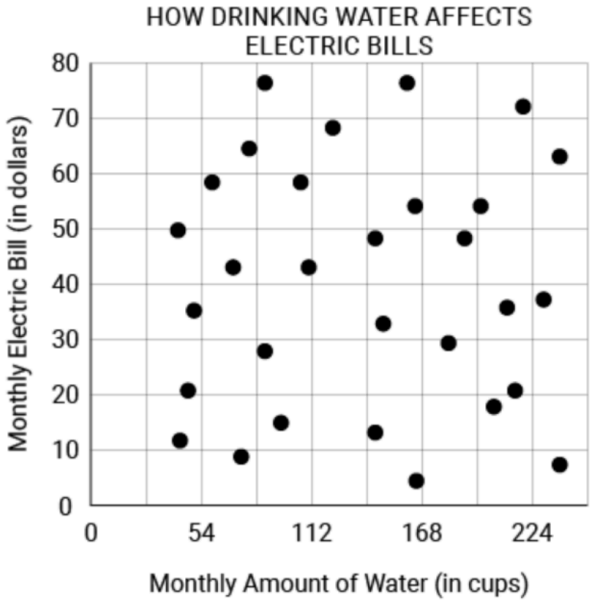
× D $(2, 1)$

7. The game store sells both board games and figurines. Board games sell for $\$40$ a game and figurines sell for $\$15$ a figurine. If the store sells 80 total pieces of merchandise and earned $\$2{,}625$ from those sales, how many board games did the game store sell?

PREV. QUESTION | NEXT QUESTION

- A. 23
- B. 40
- C. 48
- D. 57

8.



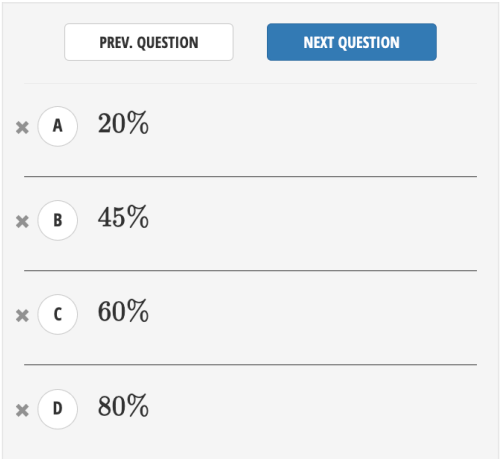
A study asked voluntary participants to record the number of cups of water they drink in a month and the electric bill for the month. The results are shown in the scatterplot. What can be inferred from the scatterplot?

PREV. QUESTION | NEXT QUESTION

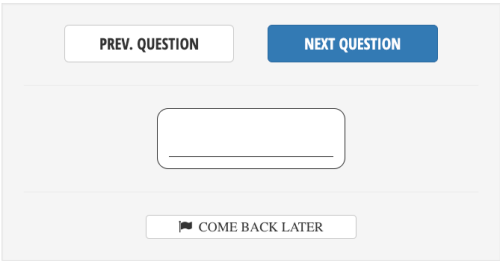
- A. No one drinks less than 27 cups of water in a month.
- B. The more water one drinks, the higher the electric bill.
- C. There is no correlation between the monthly amount of water drunk and the electric bill.
- D. Drinking bottled water explains why some electric bills are lower.

COME BACK LATER

9. A bag of candy contains a certain percentage of chocolate bars. Of the chocolate bars, 25% of them are white chocolate. If 20% of the candy in the bag is white chocolate bars, what percentage of the candy is chocolate bars?



10. The perimeter of a rectangle is 72 feet. If the length is twice the width, what is the area of the rectangle, in square feet?



11. $(x-4)^2+(y+6)^2=576$

The given equation represents a circle in the xy–plane.

What is the distance between the center of the circle and the point on the circle with the greatest value of y?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

12. In a data set, the mean is 10, the median is 10, the middle three values are 10, and the largest value is greater than 10. If the largest value is removed from the data set, which of the following correctly represents the changes to the mean and median?

PREV. QUESTION | NEXT QUESTION

A. The mean will decrease and the median will decrease.

B. The mean will decrease and the median will stay the same.

C. The mean will increase and the median will stay the same.

D. The mean will stay the same and the median will stay the same.

13. Triangle A and triangle B are similar right triangles. The hypotenuse of triangle A is 13 inches, and the hypotenuse of triangle B is 26. If the area of triangle A is 30 square inches, what is the area of triangle B, in square inches?

PREV. QUESTION | NEXT QUESTION

- A 60
- B 120
- C 180
- D 240

14. $9(2x+3)-10(x+1)=12(x-1)-2(2x+5)$

How many distinct real solutions does the given equation have?

PREV. QUESTION | NEXT QUESTION

- A Zero
- B Exactly one
- C Exactly two
- D Infinitely many

15. A line with the equation $12x - 5y = 30$ passes through the points $(0, p)$ and $(q, 0)$. What is the value of $\frac{q}{p}$?

PREV. QUESTION | NEXT QUESTION

- A $-\frac{12}{5}$
- B $-\frac{5}{12}$
- C $\frac{5}{12}$
- D $\frac{12}{5}$

16. $$\frac{(x+2)(x^2-3x-28)}{x-7}$$

The given expression is equivalent to the expression $x^2 + 2ax + 2b$. What is the value of $a + b$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

17. $$|3x - 4| = x + 12$$

The given equation has two solutions, p and q. What is the value of pq?

PREV. QUESTION | NEXT QUESTION

- A -16
- B 8
- C 16
- D 64

18. For the function $f(x) = a^x + b$, where $a > 0$, $f(1) = 12$ and $f(2) = 24$. What is the value of b?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

19. The gravity on Mercury is 3.7 meters per second squared, or $3.7 \frac{m}{s^2}$. Which of the following is equal to the gravity on Mercury in kilometers per hour squared, or $\frac{km}{h^2}$?

(1 kilometer = $1{,}000$ meters)

PREV. QUESTION | NEXT QUESTION

× A 13.32

× B $1{,}027.8$

× C $1{,}332$

× D $47{,}952$

20. Given the linear function $f(x) = ax + b$ and the quadratic function $g(x) = ax^2 + bx$, what are the values of x at which $f(x) = g(x)$?

PREV. QUESTION | NEXT QUESTION

× A $x = -1$ and $x = \frac{-a}{b}$

× B $x = -1$ and $x = \frac{-b}{a}$

× C $x = 1$ and $x = \frac{-a}{b}$

× D $x = 1$ and $x = \frac{-b}{a}$

21. The height, in meters, of a projectile launched from a cliff after t seconds can be modeled by the function $h(t) = -4.9t^2 + 19.6t + 58.8$. What is the maximum height, in meters, the projectile will reach?

PREV. QUESTION | NEXT QUESTION

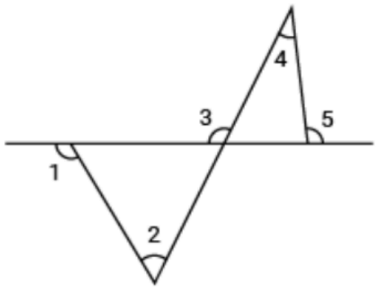
× A 2

× B 58.8

× C 78.4

× D 83.3

22.



In the figure, the measure of $\angle 1 = a^\circ$, the measure of $\angle 3 = b^\circ$, and the measure of $\angle 5 = c^\circ$. What is the sum of the measure of $\angle 2$ and the measure of $\angle 4$, in degrees?

PREV. QUESTION

A $a + b + c - 180$

B $a + 2b + c - 180$

C $a + b + c - 360$

D $a + 2b + c - 360$

COME BACK LATER